

THREAT HUNT REPORT

Unauthorized TOR Browser Installation and Use

Endpoint	virtualtest11 (IP: 10.0.0.141)
User Account	bthrasher80
Activity Window	April 25, 2026 – April 27, 2026
Platform	Microsoft Defender for Endpoint Advanced Hunting (KQL)
Analyst	Basilio Thrasher Project Basilio – SOC-Lab-Architecture
MITRE ATT&CK	T1090.003 – Proxy: Multi-hop Proxy T1486 – Data Encrypted for Impact

1. Scenario

Management suspects that employees may be using TOR browsers to bypass network security controls. Recent network logs show unusual encrypted traffic patterns and connections to known TOR entry nodes. Anonymous reports indicate employees have been discussing ways to access restricted sites during work hours. The goal of this hunt is to detect any TOR usage, analyze related security incidents, and escalate findings to management.

2. Hunt Hypotheses

Before querying any data, the following hypotheses were formed to guide the investigation:

- H1: A user on virtualtest11 downloaded and silently installed the TOR browser portable package.
- H2: The TOR process (tor.exe) was executed under a standard user account, not a system account.
- H3: The device established outbound connections to known TOR relay nodes on port 9001.
- H4: TOR established a local SOCKS proxy on 127.0.0.1, routing traffic anonymously.
- H5: Secondary malicious activity (data staging or ransomware execution) followed TOR usage.

3. Queries Used

3.1 TOR File Activity — DeviceFileEvents

Searched for any file events associated with user bthrasher80 where the filename contained 'tor'. This revealed TOR installer activity and the creation of a file named 'Tor-Shopping-List'.

```
DeviceFileEvents
| where DeviceName == "virtualtest11"
```

```
| where InitiatingProcessAccountName == "bthrasher80"  
| where FileName contains "tor"  
| order by Timestamp desc  
| project Timestamp, DeviceName, ActionType, FileName, FolderPath, SHA256,  
    Account = InitiatingProcessAccountName
```

3.2 Silent Installation — DeviceProcessEvents

Searched for process command lines referencing the TOR portable installer to detect silent installation.

```
DeviceProcessEvents  
| where DeviceName == "virtualtest11"  
| where ProcessCommandLine contains "tor-browser-windows-x86_64-portable"  
| project Timestamp, DeviceName, AccountName, ActionType, FileName,  
    FolderPath, SHA256, ProcessCommandLine
```

3.3 TOR Process Execution — DeviceProcessEvents

Searched for confirmed execution of tor.exe or tor-browser.exe under the user account, confirming the browser was actively launched.

```
DeviceProcessEvents  
| where DeviceName == "virtualtest11"  
| where FileName has_any ("tor.exe", "tor-browser.exe")  
| project Timestamp, DeviceName, AccountName, ActionType, FileName,  
    FolderPath, SHA256, ProcessCommandLine
```

3.4 TOR Network Connections — DeviceNetworkEvents (Time-Scoped)

Scoped the query to the activity window (April 25–28) to avoid baseline noise from unrelated later dates. Filtered for connections initiated by tor.exe on known TOR relay ports.

```
DeviceNetworkEvents  
| where DeviceName == "virtualtest11"  
| where Timestamp between (datetime(2026-04-25) .. datetime(2026-04-28))  
| where InitiatingProcessFileName == "tor.exe"  
| project Timestamp, RemoteIP, RemotePort, RemoteUrl, ActionType  
| order by Timestamp asc
```

3.5 pwncrypt Ransomware File Activity — DeviceFileEvents (Time-Scoped)

Searched for files containing 'pwncrypt' in the same activity window to identify ransomware staging and file encryption events.

```
DeviceFileEvents  
| where DeviceName == "virtualtest11"  
| where Timestamp between (datetime(2026-04-25) .. datetime(2026-04-28))  
| where FileName contains "pwncrypt"
```

| project Timestamp, ActionType, FileName, FolderPath, InitiatingProcessFileName

4. Chronological Event Timeline

All events below are drawn directly from MDE Advanced Hunting telemetry. Only events relevant to TOR browser usage and associated malicious activity are included.

Timestamp (UTC-5)	Event	MDE Table	Key Finding / IOC
Apr 25, 2026 07:18:20 AM	tor.exe first executed	DeviceProcessEvents	User bthrasher80 launched tor.exe from C:\Users\bthrasher80\Desktop. SHA256: a028d058c4d49cf0df10...
Apr 25, 2026 07:18:21 AM	TOR local SOCKS proxy established	DeviceNetworkEvents	ListeningConnectionCreated on 127.0.0.1:51613 and 127.0.0.1:51615 — TOR SOCKS proxy activated.
Apr 25, 2026 07:20:26 AM	First outbound TOR relay connection	DeviceNetworkEvents	ConnectionSuccess to 37.221.195.103:9001 (TOR relay port).
Apr 25, 2026 07:20:27 AM	TOR hidden service URI resolved	DeviceNetworkEvents	ConnectionSuccess to 37.221.195.103:9001 via https://www.ykawzx7s2gfsvh7iw3snm.com — confirms active TOR routing.
Apr 25, 2026 07:20:29 AM	Additional TOR relay connections	DeviceNetworkEvents	ConnectionSuccess to 141.95.86.17:9001 and 212.227.127.105:9001. Multiple .onion-style URIs resolved.
Apr 25, 2026 08:12:52 AM	pwncrypt.ps1 dropped to disk	DeviceFileEvents	FileCreated: pwncrypt.ps1 at C:\ProgramData\pwncrypt\ — initiated by powershell.exe. Ransomware script staged.
Apr 25, 2026 08:12:56 AM	File encryption begins — Round 1	DeviceFileEvents	FileCreated + FileRenamed pattern across 3 file types: 2765_CompanyFinancials_pwncrypt, 7294_ProjectList_pwncrypt, 6156_EmployeeRecords_pwncrypt. Originals moved to C:\Windows\Temp\.
Apr 25, 2026 12:12:05 PM	pwncrypt.ps1 re-executed — Round 2	DeviceFileEvents	FileCreated: pwncrypt.ps1 dropped again. Second encryption cycle begins. New encrypted filenames created across same 3 data categories.
Apr 25, 2026 04:12:58 PM	pwncrypt.ps1 re-executed — Round 3	DeviceFileEvents	Third encryption cycle. FileCreated + FileRenamed events for additional CompanyFinancials, ProjectList, and EmployeeRecords files.
Apr 25–26, 2026 (Multiple)	Sustained TOR relay activity	DeviceNetworkEvents	Continued ConnectionSuccess events to 141.95.86.17:9001 and 212.227.127.105:9001 throughout April 25–26. Multiple .onion URIs resolved across sessions.

Timestamp (UTC-5)	Event	MDE Table	Key Finding / IOC
Apr 27, 2026 11:38:10 AM	tor.exe re-executed — Session 2	DeviceProcessEvents	Second confirmed TOR browser launch. ListeningConnectionCreated on 127.0.0.1:58291. Relay connections resume to 141.95.86.17:9001 and 212.227.127.105:9001.
Apr 27, 2026 11:40:10 AM	tor.exe re-executed — Session 3	DeviceProcessEvents	Third TOR browser launch. New SOCKS proxy on 127.0.0.1:58391 and 58393. Relay connections to 141.95.86.17:9001.
Apr 27, 2026 12:28:46 PM	Tor-Shopping-List.lnk created	DeviceFileEvents	FileCreated: Tor-Shopping-List.lnk in C:\Users\bthrasher80\AppData\Roaming\Microsoft\Windows\Recent\ — user actively accessed this file via Windows Explorer. Session was a remote RDP connection from 10.0.8.9 via Guacamole RDP. SHA256: e1cfe54058e41341392af5229c01d111 ...
Apr 27, 2026 12:35:50 PM	tor.exe re-executed — Session 4	DeviceProcessEvents	Fourth and final confirmed TOR execution. Same SHA256 hash as all prior executions confirming the same binary.

5. Indicators of Compromise (IOCs)

Type	Value	Context
Process	tor.exe	Executed 4 times by bthrasher80 on virtualtest11. Path: C:\Users\bthrasher80\Desktop\
SHA256	a028d058c4d49cf0df10...	Consistent hash across all 4 tor.exe executions confirming same binary used throughout.
IP Address	37.221.195.103	TOR relay node. ConnectionSuccess on port 9001. First contact Apr 25 at 7:20 AM.
IP Address	141.95.86.17	TOR relay node. Repeated ConnectionSuccess events on port 9001 across Apr 25–27.
IP Address	212.227.127.105	TOR relay node. Repeated ConnectionSuccess events on port 9001 across Apr 25–27.
File	pwncrypt.ps1	Ransomware script dropped to C:\ProgramData\pwncrypt\ by powershell.exe. Executed 3 times.
File	Tor-Shopping-List.lnk	Windows shortcut in Recent Items folder. Created Apr 27 at 12:28 PM during RDP session from 10.0.8.9. Indicates deliberate premeditated TOR use.

6. Summary of Events

Between April 25 and April 27, 2026, user bthrasher80 on endpoint virtualtest11 deliberately installed, configured, and repeatedly used the TOR browser to anonymize network activity. The investigation uncovered a multi-day pattern of behavior that escalated from TOR browser installation to ransomware execution against company data.

On the morning of April 25, 2026, at 7:18 AM, bthrasher80 executed tor.exe from the Desktop. Within two minutes the TOR client established a local SOCKS proxy on 127.0.0.1 and began routing traffic through TOR relay nodes on port 9001, including 37.221.195.103, 141.95.86.17, and 212.227.127.105. Multiple .onion-style URIs were resolved, confirming active browsing through the TOR anonymization network.

Approximately one hour after the first TOR session, at 8:12 AM on April 25, the ransomware script pwncrypt.ps1 was dropped to C:\ProgramData\pwncrypt\ by powershell.exe and immediately began encrypting files on the Desktop. Three categories of company data were targeted: CompanyFinancials, ProjectList, and EmployeeRecords. The encryption pattern consisted of FileCreated events on the Desktop followed by FileRenamed events moving originals to C:\Windows\Temp\, effectively staging encrypted copies while removing originals. This encryption cycle ran three separate times on April 25 — at 8:12 AM, 12:12 PM, and 4:12 PM — indicating either a scheduled or manually triggered persistence mechanism.

TOR browser activity resumed on April 27, 2026, with three additional confirmed executions of tor.exe at 11:38 AM, 11:40 AM, and 12:35 PM. Each session re-established the local SOCKS proxy and reconnected to the same TOR relay nodes observed on April 25. At 12:28 PM on April 27, a Windows shortcut named Tor-Shopping-List.lnk was created in the user's Recent Items folder during a remote RDP session originating from 10.0.8.9 via Guacamole RDP. This artifact strongly suggests deliberate and premeditated intent — the user maintained a list of TOR-related activity and accessed it remotely.

In total, tor.exe was executed four confirmed times across two days. The same binary SHA256 hash was present in all four executions, confirming a single consistent TOR installation was used throughout. TOR network connections were sustained across April 25 and 26, with activity resuming on April 27.

7. Response Taken

TOR usage was confirmed on endpoint virtualtest11 under user account bthrasher80. The following actions were taken:

- Endpoint virtualtest11 was isolated from the network to prevent further data exfiltration or command-and-control activity.
- The user's direct manager was notified of the confirmed TOR usage and associated ransomware activity.
- All findings, query results, and IOCs were documented and preserved for further investigation and potential HR/legal action.
- Encrypted files (CompanyFinancials, ProjectList, EmployeeRecords variants) were flagged for recovery assessment.

8. Portfolio Note

Interview answer this exercise builds:

"I conducted a threat hunt for unauthorized TOR browser usage using Microsoft Defender for Endpoint Advanced Hunting with KQL. I queried DeviceProcessEvents, DeviceNetworkEvents, and DeviceFileEvents to reconstruct the full attack timeline — from TOR installation through active relay connections on port 9001 to pwncrypt ransomware execution — across a three-day activity window on virtualtest11, documented in the SOC-Lab-Architecture GitHub repository."

THREAT HUNT REPORT

Plain Language Explanation — TOR Browser Investigation

Endpoint	virtualtest11 (IP: 10.0.0.141)
User Account	bthrasher80
Activity Window	April 25, 2026 – April 27, 2026
Analyst	Basilio Thrasher Project Basilio – SOC-Lab-Architecture
Document Purpose	Non-technical companion to the formal Threat Hunt Report

1. What Is This Document?

This is a plain language explanation of the TOR browser threat hunt investigation conducted on endpoint virtualtest11. It is intended to accompany the formal technical report and explain what happened, how it was found, and what it means — without requiring a cybersecurity background to understand.

2. What Actually Happened — The Story

The Setup

Imagine you are a security analyst at a company. Someone in HR tips off management that employees might be using software to hide what they are doing on the internet. Management asks you to investigate.

You have access to a security tool — Microsoft Defender — that records everything that happens on company computers. Every file opened, every program run, every website visited, every connection made. It is like a security camera that never stops recording.

What Was Found

TOR Browser Was Installed and Used

The employee — bthrasher80 — installed and repeatedly used a program called TOR. TOR is a browser specifically designed to make internet activity invisible. Think of it like the difference between driving on a normal road where cameras can see you versus driving through a tunnel system with no cameras and multiple exits — nobody can track where you came from or where you are going.

Legitimate uses for TOR exist, but on a company computer it is a major red flag because it bypasses all of the company's security controls. The recordings showed TOR ran four separate times over three days — April 25th and April 27th. Each time it ran, it connected to anonymous middlemen servers located overseas and routed all traffic through them so it could not be traced.

A File Called Tor-Shopping-List Was Found

This detail changes the story from 'maybe accidental' to 'definitely intentional.' Nobody accidentally creates a shopping list for TOR activity. A Windows shortcut file named Tor-Shopping-List was found in the user's recent files, created during a remote login session. This was planned.

Ransomware Executed

About one hour after the first TOR session, a ransomware script called pwncrypt started running. Ransomware is malicious software that takes your files and locks them — like someone changing the locks on your filing cabinets and taking the keys. The script ran three times in one day and encrypted company files including financial records, employee records, and project files. The originals were moved to a hidden location and replaced with locked versions.

3. How It Was Found

The investigation followed a logical process, building each question on the answer from the last:

Step 1	Asked the system: show me any files related to TOR on this computer.
Step 2	Asked: show me any programs that ran with TOR in the name.
Step 3	Asked: show me any internet connections that went to known TOR addresses.
Step 4	Asked: show me any files created or renamed by the ransomware script.
Step 5	Noticed the results were showing the wrong time period and adjusted the search — that adjustment unlocked

4. What the Evidence Proved

In chronological order, the evidence established the following:

1	Employee installed TOR browser intentionally on a company machine.
2	Employee used TOR to anonymously browse the internet — bypassing all company security controls.

3	Employee had a premeditated plan — the Tor-Shopping-List file proves deliberate intent.
4	Ransomware (pwncrypt) executed approximately one hour after the first TOR session, encrypting company data.
5	The activity repeated across multiple days — this was not a one-time mistake.
6	All four TOR executions used the same binary file, confirming a single consistent installation throughout.

5. What This Means in the Real World

In a real company, this investigation would trigger an immediate response. The analyst's job is to find the evidence, document it clearly, and hand it off to the people who make decisions. The actions that would follow a confirmed finding like this include:

Isolation	The employee's machine would be immediately disconnected from the network to stop any further damage.
Recovery	IT would work to recover encrypted files from backups. The three file categories encrypted — financials, HR records, and legal documents.
Notification	HR, legal, and senior management would be notified of the confirmed findings and the documented evidence.
Escalation	Depending on what was accessed via TOR and what was on the shopping list, law enforcement notification would be required.
Analyst Role	The analyst's job ends at documentation. Finding the evidence, building the timeline, and presenting it to management.

6. The Interview Answer This Exercise Builds

"I conducted a threat hunt for unauthorized TOR browser usage using Microsoft Defender for Endpoint. I queried process, file, and network event logs to reconstruct the full attack timeline — from TOR installation through active connections to overseas relay nodes, to ransomware execution — across a three-day activity window. I identified the evidence, documented it chronologically, and produced a formal investigation report with IOCs and a recommended response. The full investigation is documented in my SOC-Lab-Architecture GitHub repository."